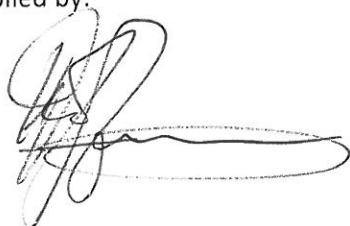
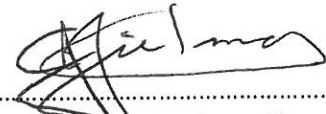
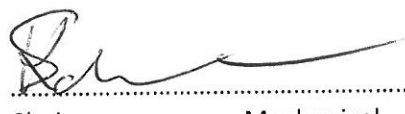


Dept. of Mechanical and Mechatronic Engineering Safety Procedure
Working at Heights (Revision 1.0)

Approval

Compiled by:  Date: 1/06/2018	Reviewed by:  Departmental safety officer Date: 1/6/18	Approved by:  Chairman: Mechanical & Mechatronic Engineering Date: 4/6/2018
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1. Revision history:

This procedure supersedes the STERG Fall Protection Plan – 18m Lattice Tower of 20 May 2013.

2. Definitions:

Effective date is the date on which this procedure was approved by the Chairman: Mechanical and Mechatronic Engineering

Inclement weather is defined as precipitation and wind speeds above 18 m/s (BS5975)

Laboratory and workshop manager as appointed by the Department

Laboratory supervisor as appointed by the Department in charge of a particular laboratory

Personal protective equipment means EN/SANS approved hard hat with chin strap, safety harness (including shock absorbing- and work positioning lanyards) and safety shoes

Post graduate coordinator as appointed by the Department

Rescuer means a permanent staff member that completed an accredited training course in fall arrest, and in possession of a certificate that falls within its expiry date

Safety officer for the workshops and laboratories of the Department

Study leader means the staff member supervising the under- or post student working on the project

The Act means the Occupational Health and Safety Act 85 of 1993

The Department means the Department of Mechanical and Mechatronic Engineering, Stellenbosch University

Trained person means a staff member or student working for the Department that completed an accredited training course in fall prevention, and is in possession of a certificate that falls within its expiry date

A **worker** refers to any student or staff member of the Department

3. Applicability:

This procedure is applicable to all staff and students of the Department. It applies to all structures without a walkway and safety railings where the feet of the worker may be more than 2 m above the ground.

This procedure only applies to on-campus facilities of the Department.

Rope access work is not allowed under this procedure.

4. Purpose

The working at heights procedure sets out the responsibilities, processes and practices to ensure the safety of workers exposed to the additional risk of working at heights.

5. Responsibilities

5.1 The Chairman responsible for annual safety induction training for all new staff and students.

5.2 Students doing work in the laboratories that includes working at heights must submit a safety plan and a copy of a valid rope climbing access certificate for approval by their study leader, laboratory supervisor and the safety officer.

5.3 The laboratory supervisors are responsible for regular inspection and maintenance of all safety equipment associated with their laboratories.

5.4 The study leaders are responsible for the safety training of all students under their supervision.

- 5.5 The Chairman of the Department is responsible for the safety training of all permanent staff.
- 5.6 The safety officer is responsible for granting access to individual laboratories.
- 5.7 The safety officer assigns a rescuer to the team to perform the work at height.
- 5.8 Workers working at height must ensure that they are medically fit to do so.

6. Procedure

- 6.1 The student must obtain written permission from the laboratory supervisor and safety officer to access the structure.
- 6.2 The student must inform the study leader and division head by e-mail at least 4 hours before they access the structure of their intention to do so.
- 6.2 No work on outdoors structures is allowed during inclement weather.
- 6.3 A pre-climb briefing is required, where the team to perform the work at height discusses the planned work to be done, emergency procedures and cross-check each other's safety equipment.
- 6.4 Control access to the area below and around the structure on which work is to be performed.
- 6.6 Workers working at height must don the appropriate personal safety equipment.
- 6.5 Support workers on the ground should wear hard hats and safety shoes.
- 6.6 Tools not in use must be safely stowed or securely tied to the workers person.
- 6.7 Tools or equipment exceeding 8 kg in mass may not be carried on the worker's person, but lifted by a winch.
- 6.8 During the work, the safety rope should be securely attached to a sturdy structure above the head of the worker where possible. Use two independent anchor points when fully suspended on equipment. When doing so, the angle subtended by the two ropes must not exceed 120°.

7. Ladders

- 7.1 Ladders may not exceed 9 m in length.
- 7.2 If a ladder is used to access a working platform, the ladder should protrude at least 1.5 m above the working platform.
- 7.3 Ladders missing anti-skid pads, rungs or pop-rivets may not be used.
- 7.4 Ladders should be adequately braced and supported, and be capable of supporting anticipated lateral and vertical loads on them.

8. Applicable documents:

- 8.1 The Occupational Health and Safety Act 85 of 1993 (OHS Act)
- 8.2 The Department of Labour's Construction Regulations, 2014
(<http://www.labour.gov.za/DOL/legislation/regulations/occupational-health-and-safety/construction-regulation-2014>)
- 8.3 Safety procedures for laboratory setups

9. Records:

- 9.1 A copy of this procedure must be posted on the Department's website.
- 9.2 The student safety plans and fall prevention training certificates are kept on record by the safety officer.
- 9.3 Staff fall prevention and/or fall arrest training certificates are kept on record by the safety officer.
- 9.4 Incident reports are kept on record by the safety officer.
- 9.5 Equipment inspection reports are kept on record by the laboratory supervisor.